

Ken Yamashiro

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Education

University of Southern California – B.S. Mechanical Engineering, GPA 3.7 Expected Graduation May 2027

Experience

USC Viterbi School of Engineering — Unmanned Aerial Systems Project December 2025 – Present

Undergraduate Assistant and Lead Student Engineer Los Angeles, CA

- Assist Dr. Penkova in characterizing flight dynamics of organic, bio-inspired aerodynamic designs on low-Reynolds-number autonomous VTOL drones in hybrid water/air applications
- Develop electrical systems, program flight dynamics, and essential systems for VTOL flight

Niagara Bottling May 2026 – August 2026

Packaging Research and Development Intern Diamond Bar, CA

- Built multi-variable regression models with ANSYS FEA outputs to predict and validate bottle lightweighting designs, cutting simulation turnaround time by 1,000+ hours for the engineering team
- Integrated simulation outputs into 60+ design iterations to create predictive load models for production

NASA Columbia Scientific Balloon Facility May 2025 – July 2025

Intern/Undergraduate Assistant Palestine, TX

- Supported the final production and payload integration of the Payload for Ultrahigh Energy Observations (PUEO) research project, a NASA Pioneers project successfully launched in December 2025
- Performed surface preparation, resolved tolerance conformities, and prepared 350+ flight components
- Effectively collaborated with over 50 researchers from 11 universities across the U.S., UK, and Taiwan

UH Department of Physics and Astronomy, High Energy Physics Group (HEPG) September 2023 – August 2025

Lab Tech II and Researcher Honolulu, HI

- Utilized Siemens Solid Edge, FEA, and drafting to design, fabricate, and test 8+ physics research projects focused on particle physics, radio frequency (RF), and space environments.
- Used CNC, CAM, manual milling, lathes, and additive manufacturing to fabricate 700+ hardware parts
- Prepared technical materials and assisted in proposal presentations to NASA, JPL, and various universities

Extracurricular Activities

USC Formula SAE Racing | Powertrain Engineer August 2025 - Present

- Designed, manufactured, and tested powertrain and cooling subsystem components for the SCR26 race car (600cc inline-4 Yamaha R6) using Siemens NX, ANSYS, and MATLAB
- Selected from 60+ members to present in the design competition at the 2026 Michigan FSAE competition
- Helped place 7th out of 110 teams by supporting assembly, physical testing, and analysis of the SCR26 car

Project Experience (More listed on GitHub Website)

Cosmic Ray Lunar Sounder (CoRaLS) NASA JPL/UH HEPG

- Fabricated and tested 8+ proprietary antenna prototypes for a lunar-ice-detection spacecraft
- Contributed to the initial mechanical design of a deployable antenna assembly that stows and unfolds to full aperture in orbit, balancing stowed volume, deployment reliability, and RF geometry

Askaryan Calorimeter Experiment (ACE) Fermilab/UH HEPG

- Fabricated and bench-tested RF waveguide prototypes used to validate a new coaxial-to-waveguide adapter design, resolving tolerances and manufacturability for a geometry unsuited to standard machining

Skills and Certifications

Skills: SolidWorks, Siemens NX, ANSYS, FEA analysis, MiniTab, Manual/CNC Machining, GD&T, 3D printing, MATLAB

Experience in: Additive/Subtractive manufacturing, 6061, 7075 Al, Brass 360, CF tubes/honeycomb, 100/101 Cu, more

SOLIDWORKS CAD Design Associate Certification, Certificate ID: C-SY4HKFETCM